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Color image encryption based on discrete trinion Fourier transform and compressive sensing

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Abstract
To protect visual content in color images, this paper presents an encryption method by employing discrete trinion Fourier transform and compressive sensing. Firstly, each color image is precoded into a trinion matrix for holistically processing. The discrete trinion Fourier transform is performed to construct an enducing matrix. With measurement matrices generated by 3D Lorenz chaotic map, complex-type compressed sensing is then utilized to obtain sparse coefficients. Followed by Josephus scrambling and inverse discrete trinion Fourier transform, the ciphertext can be obtained. The plain-text images can be restored perfectly and the maximum PSNR value is more than 280dB. Additionally, the proposed cryptosystem shows high-level sensitivity, which is evidenced by NPCR and UACI values beyond 99.9998% and 33.3337%, respectively. Compared with several existing methods, the proposed algorithm exhibits stronger resistance against Gaussian noise.

Keywords Image encryption · Discrete trinion Fourier transform · 3D Lorenz chaotic map · Compressive sensing

1 Introduction

With the advent of the era of massive data, a large amount of multimedia especially digital images are transmitted and shared over the Internet. In particular, the content of digital images implicit some privacy [1] or sensitive information [2], and how to secure the dissemination and usage of digital contents has become an important concern [3, 4].

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Digital image encryption serves as an effective technology to protect visual content by transforming meaningful images into an unreadable ciphertext through spatial or frequency-domain strategies. Importantly, the original image can be restored using the authorized key. To name a few, Hua et al. [5] introduced an encryption scheme using Josephus problem and filtering diffusion. Hosny et al. [6] combined a 6D hyperchaotic system with Fibonacci Q-matrix (HSFQM) to encrypt images, utilizing the former for confusion and the latter for diffusion. Similarily, another color image encryption method based on fractional-order hyperchaotic system was presented in [7]. Kaur et al. [8] devised lightweight biomedical image encryption technology by integrating 7D hyperchaotic mapping and permutation strategies, enabling real-time encryption of limited image data. Gao et al. [9] employed a plane superposition principle to divide and superimpose multiple images into a cube, followed by DNA encoding, addition and mutation for multi-image encryption. However, its computational efficiency needs to be further improved. In [10], a color image encryption method utilizing blocks scrambling and chaos (BS-Chaos) was proposed, where the logistic map was employed to diffuse the scrambled image. Hosny et al. [11] combined multichannel fractional orthogonal Gegenbauer moments (FrMGMs) with a 2D logistic-sine map to encrypt color images. Initially, the color images were translated into FrMGM coefficients, which were subsequently encrypted using the turbulent dynamics of the 2D logistic-sine map. However, these image encryption algorithms in the spatial domain are susceptible to attacks.

To enhance the robustness, the cryptosystems can be combined with discrete Fourier transform, gyrator transform and others. For instance, Shao et al. [12] introduced a double image encryption method based on two-dimensional discrete Fourier transform (2D-DFT) and equal modulus decomposition, where the conjugate symmetry property of 2D-DFT was skillfully used. Ren et al. [13] employed phase truncation to asymmetrically encrypt images through discrete multi-parameter fractional Fourier transform. In [14], chaotic map was combined with mixture amplitude-phase retrieval to achieve real-valued image encryption, the disturbed plaintext images were combined into a complex matrix, and then the amplitude and phase were iteratively restored in gyrator transform domain. Yu et al. [15] proposed a four-image encryption scheme utilizing quaternion Fresnel transform and computer-generated hologram, which effectively processed complex value images and enhanced the security of cryptosystem.

By using discrete trinion Fourier transform and compressive sensing, we investigate a novel multiple color images encryption algorithm. The main contributions are summarized as: (1) An encryption method for color images based on trinion transform is constructed, it is more robustness compared with the spatial encryption methods. (2) The enducing matrix of discrete trinion Fourier transform is given, it can be converted to the plural form for further processing. (3) The initial value of chaotic sequence depends on the plaintext image, which ensures the system has high security together with Joseph scrambling.

The remainder of the paper is organized as follows. Some related work is briefly reviewed in Section 2. Section 3 introduces preliminaries used in the proposed scheme. The encryption and decryption processes of the proposed algorithm are elaborated in Section 4. Numerical simulations are provided in Section 5 and Section 6 concludes this work.

2 Related work

Compared with single-image encryption system, multi-image encryption system can transmit more data simultaneously, which is very important in practical applications. Pan et al. [16] proposed a multi-image encryption algorithm by jointing nonlinear fractional

Mellin transform with discrete cosine transform, which could achieve a trade-off between storage capacity and the quality of decrypted images. To reduce the amount of transmitted data, the compressive sensing (CS) based color image encryption was developed in [17]. Chen et al. designed an encryption scheme integrated compressed sensing, equal modulus decomposition and discrete fractional random transform (CS-EMD-DFrT) [18], which achieved encryption and compression simultaneously. Ye et al. [19] proposed a double image encryption system that employed compressed sensing and elliptic curves techniques, where the experimental results demonstrated that the quality of the decrypted images was not perfect. In [20], three gray images were simultaneously encrypted and embedded into a color carrier image, thereby boosting transmission efficiency. But the whole process was extremely complicated. Sui et al. developed an interference based multiple image encryption scheme, which employed the cascaded fractional Fourier transform (CFrFT) [21]. However, with the increase of layers, the accumulated error will degrade the quality of restored image.

Recently, multiple color images encryption technique has attracted much attention [22–27]. Joshi et al. [22] proposed to convert triple color images into a Bayer image, which was followed a multi-parameter fractional discrete Fourier transform and 3D Arnold scrambling to obtain ciphertext. Aburub and Alfalou [26] proposed an encryption method for multiple color images, which followed by the application of compressed sensing and fractional Fourier transform. Shao et al. developed a cryptosystem for multiple color images using equal module decomposition in quaternion gyrator domain [27]. However, the encrypted result is a full quaternion matrix, which increases the storage cost.

Compared with the quaternion matrix representation, the trinion matrix can encode three color components of a color image as a whole, remaining three components after different operations [28]. Yao et al. [29] developed a color image encryption method by using discrete trinion Fourier transform, where the real matrix representation makes the size of the ciphertext much larger than that of the plaintext. The same strategy was used in [30], where multiple color images were encrypted simultaneously. Recently, trinion discrete cosine transform has been introduced and utilized for a single color image encryption [31]. In this paper, we focus on exploring trinion matrix representation for multiple color images encryption.

3 Preliminaries

3.1 3D Lorenz chaotic system

Mathematically, 3D Lorenz chaotic system [32] is defined as,

$$\begin{cases} \dot{x} = -r(x - y) \\ \dot{y} = px - y - xz \\ \dot{z} = xy - bz \end{cases} \tag{1}$$

where r, b, $p \in R^+$ are the system parameters, the system is a typical two-wing chaotic attractor when $r = 10$, $b = 8/3$, $p = 28$.

3.2 Trinion and discrete trinion Fourier transform

A trinion number is represented as,

$$t = a_0 + ia_1 + ja_2 \tag{2}$$

where $\{a_0, a_1, a_2\}$ are real numbers, $\{i, j\}$ are imaginary parts satisfying the rules,

$$i^2 = j, ij = ji = -1, j^2 = -i \tag{3}$$

Let $f_t(m,n)$ be a trinion matrix with size of $M \times N$, the discrete trinion Fourier transform (DTFT) is given as [29, 30],

$$T_\theta(u,v) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f_t(m,n)\left\{\cos\left[2\pi\left(\frac{um}{M}+\frac{vn}{N}\right)\right] - \mu_{1,\theta}\sin\left[2\pi\left(\frac{um}{M}+\frac{vn}{N}\right)\right]\right\} \tag{4}$$

and the inverse discrete trinion Fourier transform (IDTFT) can be written as,

$$\hat{f}(m,n) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} T_\theta(u,v)\left\{\cos\left[2\pi\left(\frac{um}{M}+\frac{vn}{N}\right)\right] + \mu_{2,\theta}\sin\left[2\pi\left(\frac{um}{M}+\frac{vn}{N}\right)\right]\right\} \tag{5}$$

where $\mu_{1,\theta} = i\cos\theta + j\sin\theta$ ($\theta \in [0, 2\pi]$) and $\mu_{2,\theta}$ is a trinion number satisfying,

$$\mu_{2,\theta} = \frac{\sin\theta\cos\theta}{\cos^3\theta - \sin^3\theta} - i\frac{\sin^2\theta}{\cos^3\theta - \sin^3\theta} + j\frac{\cos^2\theta}{\cos^3\theta - \sin^3\theta} \tag{6}$$

Obviously, it is not possible for the values of θ to be $\{\pi/4, 5\pi/4\}$.

3.3 Josephus traversal

Josephus scrambling is a looping traversal problem [5]. Consider a circular arrangement of S elements, where d represents the starting position. Starting from the position d, loop through them in turn and remove the i-th element using stride L_i, then continue this process from the next element until the last one is removed. Mathematically, it can be expressed as: $seq = J(S, L, d)$.

As shown in Fig. 1, suppose that $S = 9$, $L = [2,3,1,4,3,2,3,1]$ and $d = 1$. By arranging elements {225,61,128,238,122,155,78,32,246} in a circle and starting from the first element, a stride length of 2 is utilized to remove the second element 61. Subsequently, starting from the

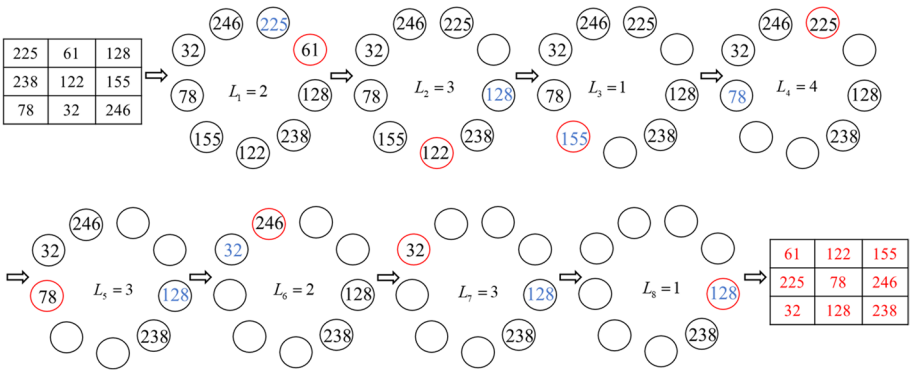


Fig. 1 The illustration of Josephus traversal

third element, a stride length of 3 is adopted to remove the fifth element 122. Continue this process until none remain, the resulting sequence is {61,122,155,225,78,246,32,128,238}.

3.4 Compressed sensing

The theory of compressed sensing is commonly employed for sparse sampling and reconstruction of original signal. The resolution of the compressed sensing problem within the complex domain can be described as:

$$\begin{aligned} &\min_{x\in C^n} \|x\|_0 \\ &s.t. Ax = y \end{aligned} \tag{7}$$

where $\|x\|_0$ is the number of non-zero elements in the vector x, $A \in C^{m\times n}$ is the sensing matrix, $y \in C^m$ is the sampling amount, $x \in C^n$ is the vector to be restored.

In this paper, the smooth l_0 (SL0) [33] reconstruction algorithm is adopted.

4 Proposed scheme

4.1 The encryption process

The framework of encryption process is shown in Fig. 2, which consists of discrete trinion Fourier transform, construction of complex matrices, compressed sensing and Josephus traversal. Let $f_l(x,y)$ $(l=1,\ 2,\ 3)$ be color images with the same size of $M\times N\times 3$ to be encrypted, the detailed encryption process is depicted as,

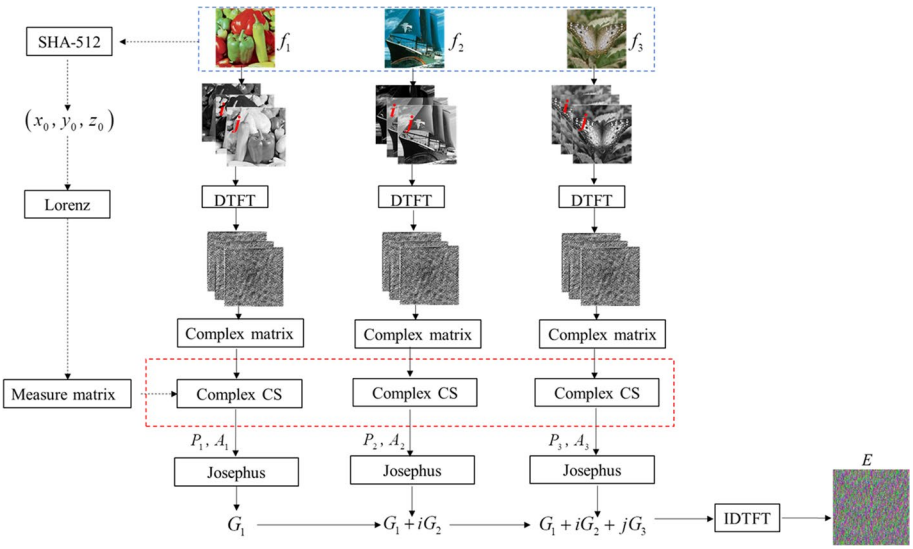


Fig. 2 Flowchart of the encryption process

(1) Firstly, the sum of three plaintext images is computed. We use SHA-512 to get hash value and it is converted into decimal to obtain $K^{'}$,

$$K' = \left\{k_1', k_2', \cdots, k_{64}'\right\} \tag{8}$$

Then, three values $\{x_0, y_0, z_0\}$ are calculated as,

$$\begin{cases} x_0 = \mathrm{mod}\left(\sum_{n=1}^{42} k_n^{'}/256, 1\right) \\ y_0 = \mathrm{mod}\left(\sum_{n=43}^{55} k_n^{'}/256, 1\right) \\ z_0 = \mathrm{mod}\left(\sum_{n=56}^{64} k_n^{'}/256, 1\right) \end{cases} \tag{9}$$

According to the Eq. (1), the chaotic sequences (X, Y, Z) with the length of $4MN$ are generated, which are used to construct the measurement matrix.

(2) Each plaintext image $f_l(x, y)$ is divided into the red channel $f_{l,\,r}(x, y)$, the green channel $f_{l,\,g}(x, y)$ and the blue channel $f_{l,\,b}(x, y)$, which are represented into trinion matrices as,

$$f_{l,t}(x, y) = f_{l,r}(x, y) + if_{l,g}(x, y) + jf_{l,b}(x, y) \tag{10}$$

Then the DTFT with different θ_l is performed to obtain matrices $T_l(u, v)$

$$T_l = DTFT\left(f_{lt}, \theta_l\right); l = 1, 2, 3 \tag{11}$$

(3) For the transformed results $T_l(u, v)$, we use operators $s\{\cdot\}, x\{\cdot\}, y\{\cdot\}$ to extract the real and imaginary parts. Since the size of the real matrix corresponding to trinion matrix is too large, we develop the enducing matrices C_l as,

$$C_l(u, v) = \begin{bmatrix} s\left\{T_l(u, v)\right\} & -\left(x\left\{T_l(u, v)\right\} + iy\left\{T_l(u, v)\right\}\right) \\ x\left\{T_l(u, v)\right\} - iy\left\{T_l(u, v)\right\} & s\left\{T_l(u, v)\right\} \end{bmatrix}; l = 1, 2, 3 \tag{12}$$

It should be noticed that the size of C_l is $2M \times 2N$.

(4) To enhance the randomness of chaotic sequence, the sequences (X, Y, Z) are randomly disarranged, and then rearranged in columns to form the matrices $\{X^{'}, Y^{'}, Z^{'}\}$ with the size of $2M \times 2N$. The measurement matrices $\{M_1, M_2, M_3\}$ are formed as,

$$\begin{cases} M_1 = \sqrt{\frac{1}{M}} X' + j\sqrt{\frac{1}{M}} X' \\ M_2 = \sqrt{\frac{1}{M}} Y' + j\sqrt{\frac{1}{M}} Y' \\ M_3 = \sqrt{\frac{1}{M}} Z' + j\sqrt{\frac{1}{M}} Z' \end{cases} \tag{13}$$

where $\sqrt{\frac{1}{M}}$ serves as a normalized coefficient, which increases the stability of the measurement matrix and reduces the reconstruction errors.

By using the SL_0 decomposition approach in [33], we obtain three sparse matrices $C_l^{'}$. The amplitude and phase matrices are expressed as,

$$C_l' = C_l \times M_l; l = 1, 2, 3 \tag{14}$$

$$A_l = PT\left\{C_l'\right\}, P_l = AT\left\{C_l'\right\}; l = 1, 2, 3 \tag{15}$$

where the operators $PT\{\cdot\}$ and $AT\{\cdot\}$ represent the phase and amplitude truncation separately.

(5) The amplitude matrices A_l are confused by using Josephus traversal with stride s_l to get G_l, which are encoded into a trinion matrix as,

$$G = G_1 + iG_2 + jG_3 \tag{16}$$

By performing the inverse DTFT with θ_4, we get the matrix $E_t(x^{'}, y^{'})$,

$$E_t = IDTFT\left(G, \theta_4\right) \tag{17}$$

The final ciphertext $E(x^{'}, y^{'})$ is the concatenation of three parts $\{s(E_t), x(E_t), y(E_t)\}$.

4.2 The decryption process

With the granted keys, the plaintext images can be recovered by decryption operation. The flowchart of the decrypted process is shown in Fig. 3 and the specific steps include,

(1) Three channels of ciphertext image $E(x^{'}, y^{'})$ are represented into a trinion matrix. Followed by the DTFT, we get matrix $\hat{E}\left(x', y'\right)$. Then one real and two imaginary parts $\left\{\hat{G}_1, \hat{G}_2, \hat{G}_3\right\}$ are obtained as,

$$\begin{cases} \hat{G}_1\left(x', y'\right) = s\left\{\hat{E}\left(x', y'\right)\right\} \\ \hat{G}_2\left(x', y'\right) = x\left\{\hat{E}\left(x', y'\right)\right\} \\ \hat{G}_3\left(x', y'\right) = y\left\{\hat{E}\left(x', y'\right)\right\} \end{cases} \tag{18}$$

(2) With the stride s_l, the matrices $\hat{G}_l$ are performed inverse Josephus traversal to acquire matrices $\hat{A}_l$. Then complex matrices $\hat{C}_l'$ can be constructed as,

$$\hat{C}_l' = \hat{A}_l \exp\left(jP_l\right); l = 1, 2, 3 \tag{19}$$

(3) By using the SL0 reconstruction algorithm to recover matrices $\hat{C}_l$ from matrices $\hat{C}_l'$.

(4) The matrices $\hat{C}_l$ are divided into 2×2 non-overlapping blocks, the matrices $\hat{T}_l$ are computed as,

$$\hat{C}_l = \begin{pmatrix} \hat{C}_{l1} & \hat{C}_{l2} \\ \hat{C}_{l3} & \hat{C}_{l4} \end{pmatrix}; l = 1, 2, 3 \tag{20}$$

$$\begin{cases} \hat{T}_{lr} = \left(\hat{C}_{l1} + \hat{C}_{l4}\right)/2 \\ \hat{T}_{lg} = \left(\mathrm{Re}\left(\hat{C}_{l3}\right) - \mathrm{Re}\left(\hat{C}_{l2}\right)\right)/2 \\ \hat{T}_{lb} = \left(-\mathrm{Im}\left(\hat{C}_{l2}\right) - \mathrm{Im}\left(\hat{C}_{l3}\right)\right)/2 \end{cases} ; l = 1, 2, 3 \tag{21}$$

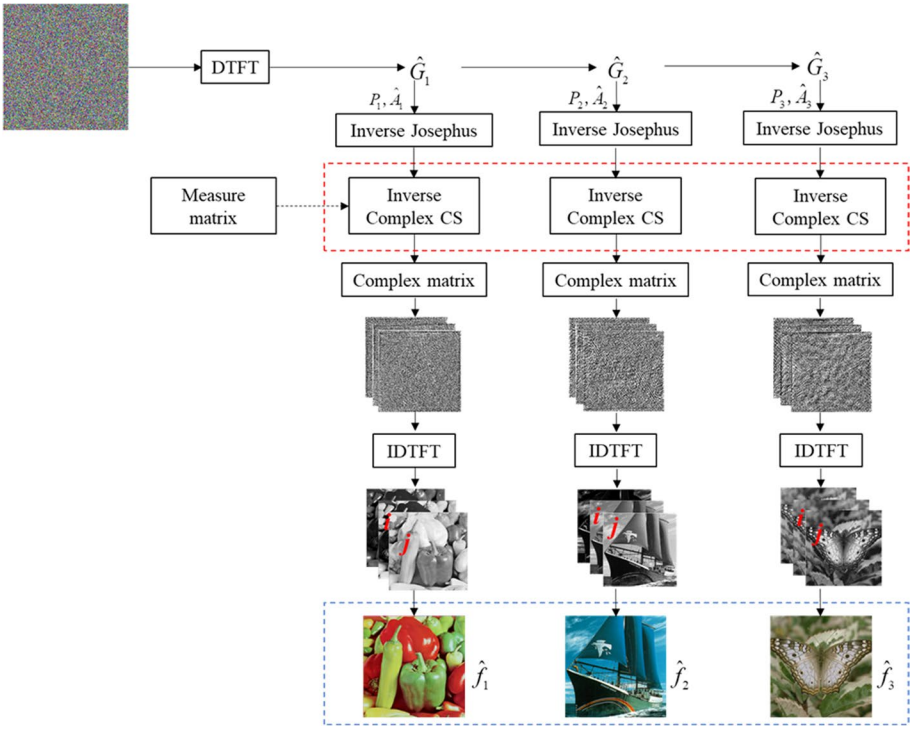


Fig. 3 Flowchart of the decryption process

where the symbols Re(·) and Im(·) denote to extract the real and imaginary part of the complex-value matrix.

(5) The inverse DTFT is conducted to acquire matrices $\hat{f}_{l,t}$, the retrieved color images are expressed as,

$$\hat{f}_l(x,y) = \left[s\{\hat{f}_{l,t}(x,y)\}, x\{\hat{f}_{l,t}(x,y)\}, y\{\hat{f}_{l,t}(x,y)\}\right]; l = 1,2,3 \tag{22}$$

5 Experimental results

In this section, numerical simulations are performed to evaluate the feasibility and validity of the proposed algorithm. A totally 300 color images selected from UGR (http://decsai.ugr.es/cvg/dbimagenes/), GS CH Database [34] and DIV2K Database [35] are used, which are resized to 256×256. Three color images are shown in Fig. 4, and the parameter values are listed in Table 1, where the θ values in DTFT and strides in Josephus are randomly chosen.



(a) (b) (c)

Fig. 4 Three tested images : (**a**) peppers, (**b**) grnpeace, (**c**) butrfly1

5.1 Results of NIST SP 800–22

To confirm the randomness of Lorenz chaotic system under the given parameters (r, b, p) in subsection 3.1, Table 2 presents the results of NIST SP 800-22 in which all the P (Probability) values are greater than 0.01. In other words, all the 15 tests passed, indicating that the Lorenz chaotic system generates random numbers with the above parameter values.

5.2 Encryption and Decryption results

The encrypted and correctly decrypted results are shown in Fig. 5, it can be clearly seen that there seems nothing understandable in ciphertext, while no significant visual difference can be discovered between the correctly decrypted results and plaintext images.

To quantitatively evaluate the quality of decrypted images, we compute Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Metric (SSIM) as,

$$PSNR = 10\log_{10}\left(\frac{255^2}{\frac{1}{3NM}\sum_{y=0}^{N-1}\sum_{x=0}^{M-1}\left(f(x,y)-\hat{f}(x,y)\right)^2}\right) \tag{23}$$

$$SSIM = \frac{\left(2\mu_f\mu_{\hat{f}}+c_1\right)\left(2\sigma_{\hat{f}f}+c_2\right)}{\left(\mu_f^2+\mu_{\hat{f}}^2+c_1\right)\left(\sigma_f^2+\sigma_{\hat{f}}^2+c_2\right)} \tag{24}$$

Table 1 The used parameters in experiments

Parameters		Values
DTFT	$(\theta_1,\theta_2,\theta_3)$	(0.2611,1.8718,1.1833)
IDTFT	θ_4	2.9500
Lorenz	(x_0,y_0,z_0)	(0.3008, 0.9258,0.8672)
Josephus	(s_1,s_2,s_3)	(4, 4, 4)

Table 2 Results of NIST SP 800-22

Test Items	*P*-value	Result
Frequency	0.9643	Pass
Block Frequency	0.6371	Pass
Cumulative Sums	0.2757	Pass
Runs	0.4373	Pass
Longest Run	0.1626	Pass
Rank	0.5341	Pass
FFT	0.4372	Pass
Non-overlapping Template	0.1223	Pass
Overlapping Template	0.5341	Pass
Universal	0.3505	Pass
Approximate Entropy	0.7399	Pass
Random Excursions	0.0487	Pass
Random Excursions Variant	0.8343	Pass
Serial	0.6371	Pass
Linear Complexity	0.8343	Pass

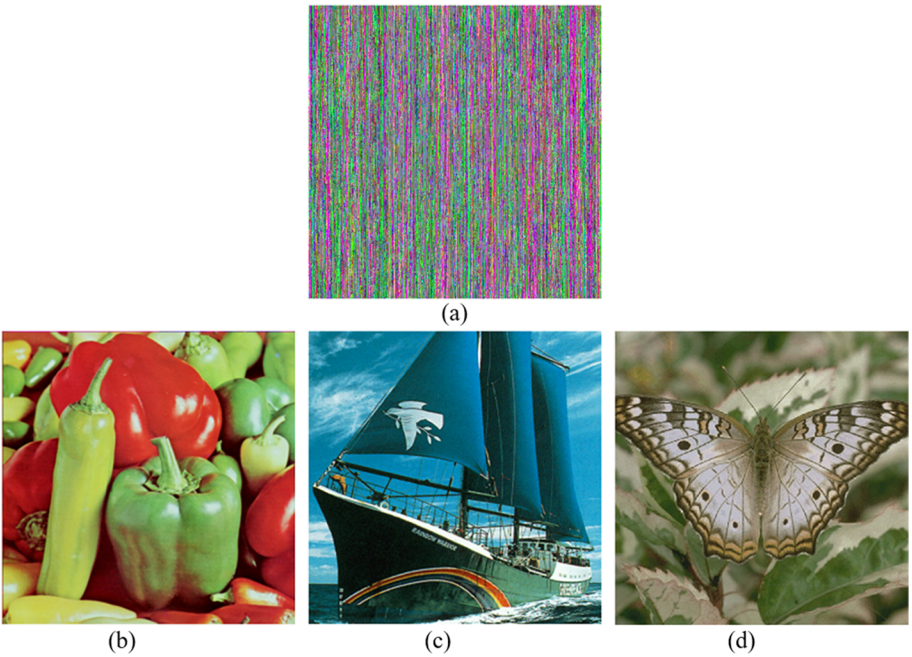


Fig. 5 Results of encryption and decryption: (**a**) ciphertext image, (**b**) the decrypted peppers, (**c**) the decrypted grnpeace, (**d**) the decrypted butrfly1

Table 3 Statistical results of PSNR and SSIM values

Methods	PSNR/dB			SSIM
	Maximum	Minimum	Mean	
The HSFQM [6]	N/A	N/A	N/A	1.0000
The BS-Chaos [10]	N/A	N/A	N/A	1.0000
The CS [17]	241.7095	152.3783	212.0466	1.0000
The CS-EMD-DFrT [18]	251.8131	171.7112	232.6535	1.0000
The CFrFT [21]	178.7877	174.4538	176.9592	1.0000
The proposed	280.1232	251.0097	261.4268	1.0000

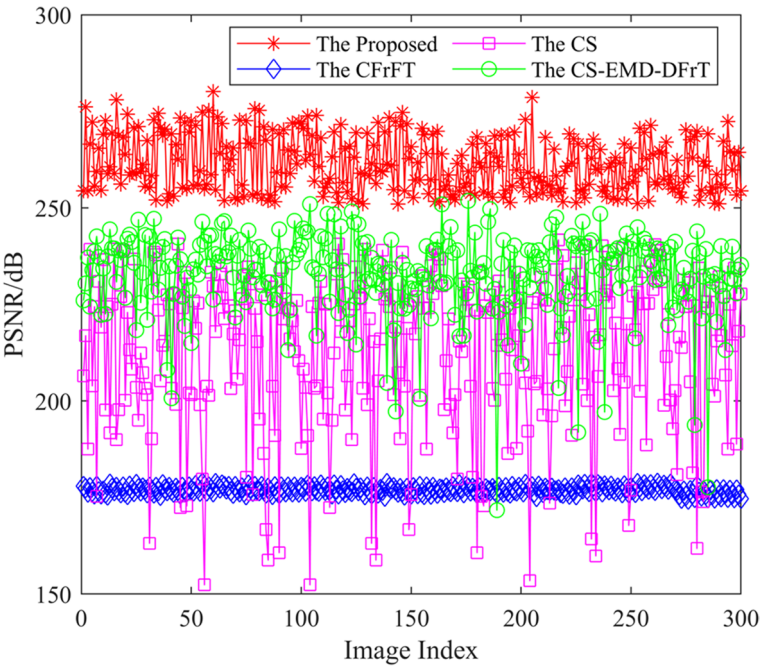


Fig. 6 Comparison results of all PSNR values

where $f(x,y)$ and $\hat{f}(x,y)$ represent the plaintext image and decrypted result respectively. $\{\mu_f, \mu_{\hat{f}}\}$ are mean values, $\{\sigma_f, \sigma_{\hat{f}}\}$ are standard deviations,$\sigma_{f\hat{f}}$ is covariance.

The decrypted results of the proposed scheme are compared with those by using the HSFQM method [6], the BS-Chaos method [10], the CS method [17], the CS-EMD-DFrT method [18] and the CFrFT method [21]. Table 3 lists the statistical results of PSNR and SSIM, where the PSNR values of spatial encryption methods in [6, 10] are the best, while the PSNR values of the proposal are between 251.0097dB and 280.1232dB, which are larger than those by using the methods in [17, 18, 21]. The SSIM values of all the methods are equal to 1.0000. The PSNR values of all the decrypted images shown in Fig. 6 further demonstrate that the proposed method can recover the original image well.

5.3 Statistical test

In this subsection, histogram and the correlation of adjacent pixels are tested. Fig. 7 shows the histogram results of the plaintext images in Fig. 4 and the ciphertext image in Fig. 5a, where the pixel distribution of plaintext images and ciphertext images are obviously different. The histogram of ciphertext is relatively concentrated, showing a Gaussian distribution. The results demonstrate that the proposed algorithm destroys the pixel distribution of original plaintext images.

Usually, there is a strong correlation between two adjacent pixels of a plaintext image. Moreover, the encryption will destroy this behavior. The correlation coefficients in horizontal, vertical and diagonal directions are calculated as,

$$\begin{cases} r_{xy} = \dfrac{E\{[x-E(x)][y-E(y)]\}}{\sqrt{D(x)}\sqrt{D(y)}} \\ E = \dfrac{1}{N}\sum_{i=1}^{N} x_i \\ D(x) = \dfrac{1}{N}\sum_{i=1}^{N}\left(x_i - E(x)\right)^2 \end{cases} \tag{25}$$

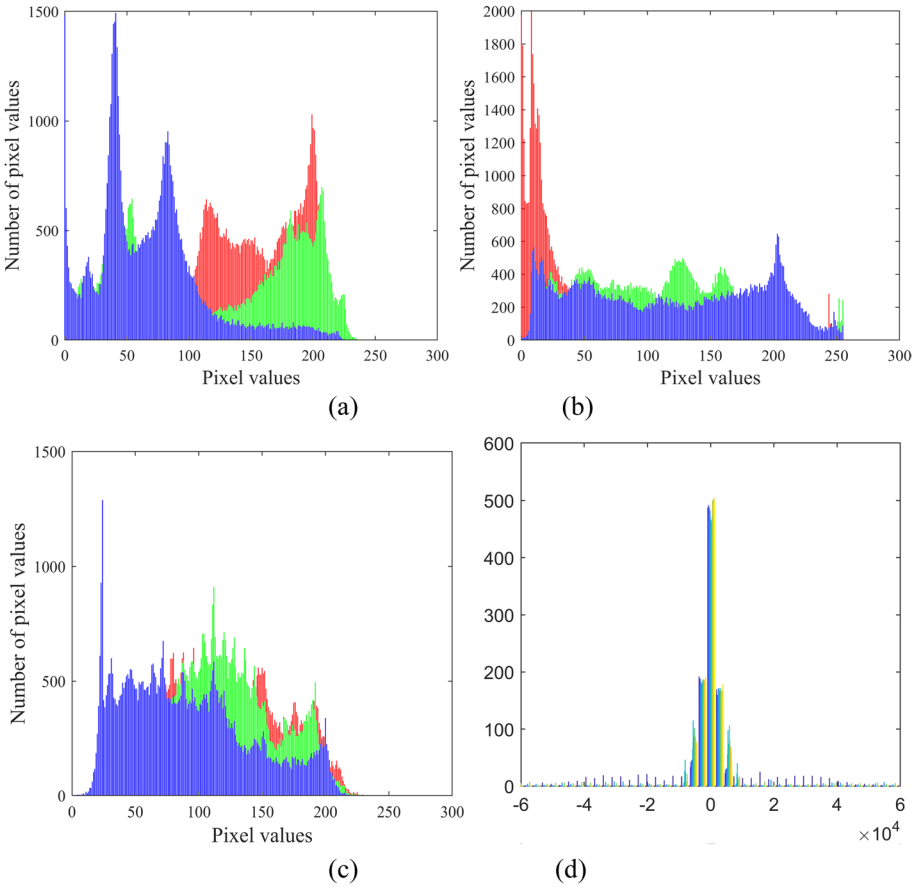


Fig. 7 The histograms: (**a**)-(**c**) are results of Fig. 4a-c, (**d**) is result of Fig. 5a

In the experiment, 5000 pairs of adjacent pixels are randomly selected from the plaintext and ciphertext images respectively. From the result list in Table 4 and shown in Fig. 8, it can be seen that the CC values of three channels in plaintext images are close to 1.0000, which is very contrary to that of ciphertext image. This indicates that the proposed algorithm meets expectations.

5.4 Key sensitivity analysis

In general, when the key space is sufficiently large, the cryptographic system exhibits higher security. Moreover, even a slightly altered correct authorization key should not contain any information about the plaintext image in the decrypted image. The algorithm utilizes 3D Lorenz chaotic mapping as the primary key space, and assumes that the parameters and initial values in 3D Lorenz chaotic system deviate from the correct value by 10^{-18}.

Fig. 9 depicts the PSNR values for incorrect decryption results, where the maximum PSNR value is less than 0. It further proves that nothing useful can be obtained when any key changes slightly. With such precision, the key space is at least up to 2^{358}, which demonstrates that the proposed algorithm can resist the brute force attack effectively.

Next, the plaintext sensitivity of the proposed algorithm is analyzed. When the plaintext image changes slightly, the new ciphertext should be significantly different from the original ciphertext. Here, the number of pixels change rate (NPCR) and unified average changing intensity (UACI) are selected as criterion and computed as,

$$NPCR = \frac{1}{NM}\sum_{j=0}^{N-1}\sum_{i=0}^{M-1} D(i,j) \times 100\% \tag{26}$$

$$UACI = \frac{1}{NM}\sum_{j=0}^{N-1}\sum_{i=0}^{M-1} \frac{|C_1 - C_2|}{F} \times 100\% \tag{27}$$

Table 4 CC values of two adjacent pixels in three directions

		Horizontal	Vertical	Diagonal
Fig. 4a	R	0.9664	0.9645	0.9393
	G	0.9751	0.9700	0.9467
	B	0.9643	0.9554	0.9275
Fig. 4b	R	0.8531	0.9267	0.8329
	G	0.8610	0.9177	0.8359
	B	0.8956	0.9219	0.8658
Fig. 4c	R	0.8792	0.9084	0.8495
	G	0.8641	0.8933	0.8170
	B	0.8920	0.9135	0.8523
Fig. 5a	R	-0.4569	0.0253	-0.0189
	G	-0.7969	-0.0148	0.0203
	B	-0.6789	0.0317	-0.0318

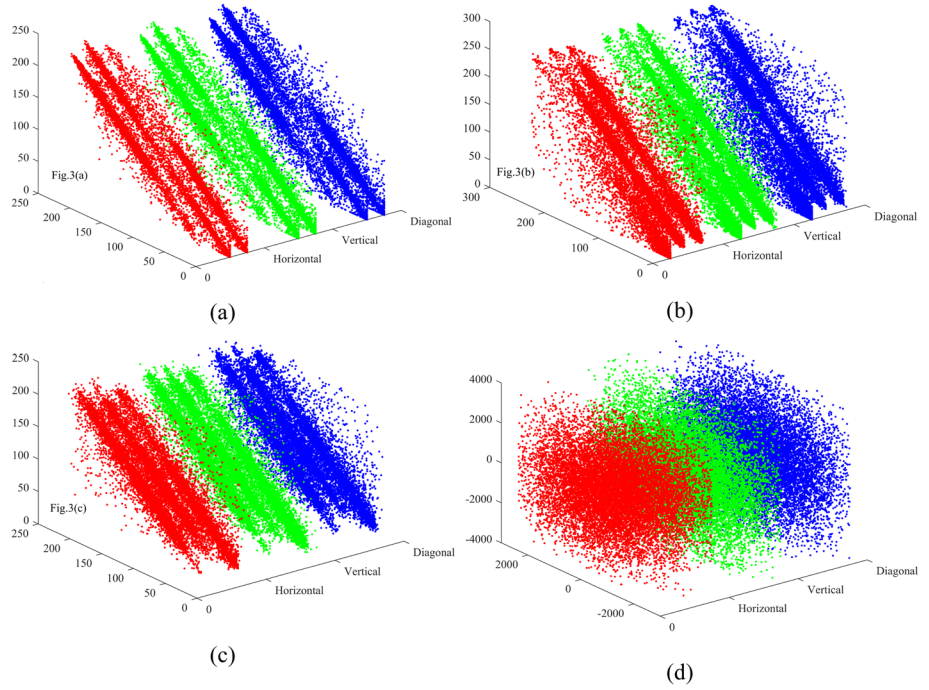


Fig. 8 Adjacent pixel pairs along the horizontal, vertical, and diagonal directions: (**a**)-(**c**) are results of Fig. 4a-c; (**d**) is result of Fig. 5a

$$D(i,j) = \begin{cases} 1, & C_1(i,j) \neq C_2(i,j) \\ 0, & C_1(i,j) = C_2(i,j) \end{cases} \tag{28}$$

where $C_1(i,j)$ is ciphertext generated from original plaintext, $C_2(i,j)$ is ciphertext generated from plaintext with small change, and F represents the maximum value in the ciphertexts.

In the experiment, one pixel of the plaintext image is randomly selected, and its pixel value is increased by 20. From the average results of 50 tests shown in Fig. 10, we can see that the NPCR and UACI are respectively greater than 99.9998% and 33.3337%, which demonstrates that the proposed algorithm has better plaintext sensitivity.

5.5 Chosen-plaintext attack

To test the resistance the proposed algorithm against chosen-plaintext attack, three pseudo-plaintext images randomly selected from the UGR database were utilized for decryption. Suppose the attacker uses the pseudo-plaintext to generate the corresponding decryption key and decrypt Fig. 5a, the results are presented in Fig. 11d-f. Obviously, the attacker cannot obtain any information about the original plaintext image. Since each encryption uses plaintext pixels to iterative to generate a chaotic sequence, the keys obtained by different

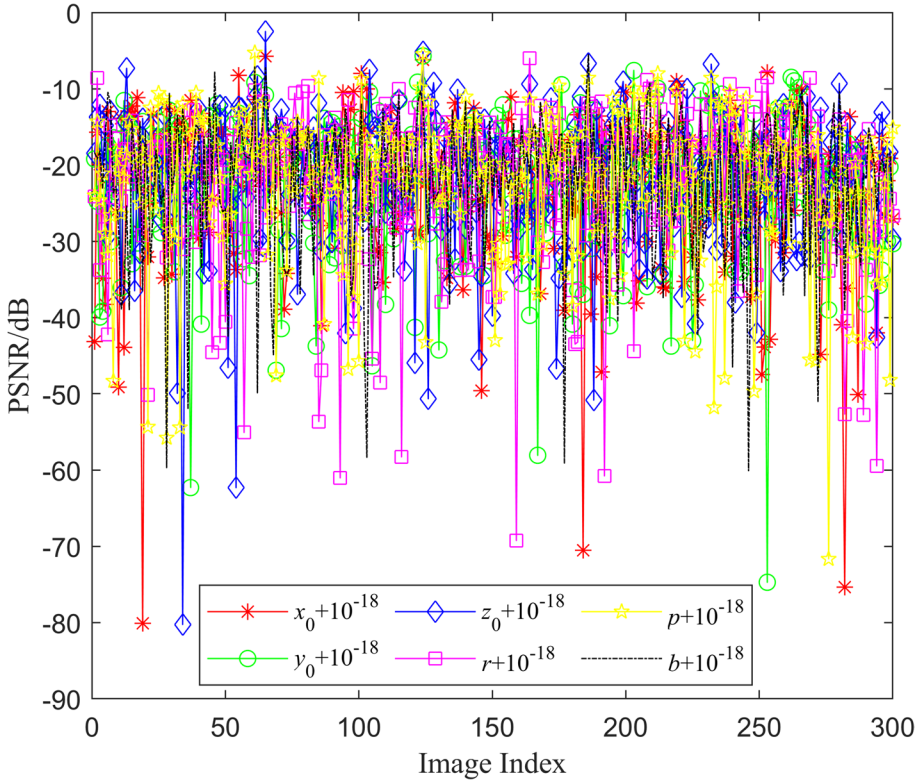


Fig. 9 The PSNR results when parameters and initial values of Lorenz chaotic system change slightly

plaintext encryptions are different. The results demonstrate that the proposed algorithm has strong resistance to chosen-plaintext attack.

5.6 Complexity analysis

When three color images are used for encryption, Fig. 12 shows the times consumption of sequence generation, discrete trinion Fourier transform, complex matrix construction, confusion, inverse discrete trinion Fourier transform and complex-type compressive sensing. Among them, the CS operation is the most time-consuming. This is mainly because it is an iterative optimization process.

5.7 Comparison with other methods

To test the robustness of the proposed algorithm, Gaussian random noise with zero mean and different intensities is added to the ciphertext, where the results are compared with those by using the HSFQM method [6], the BS-Chaos method [10], the CS

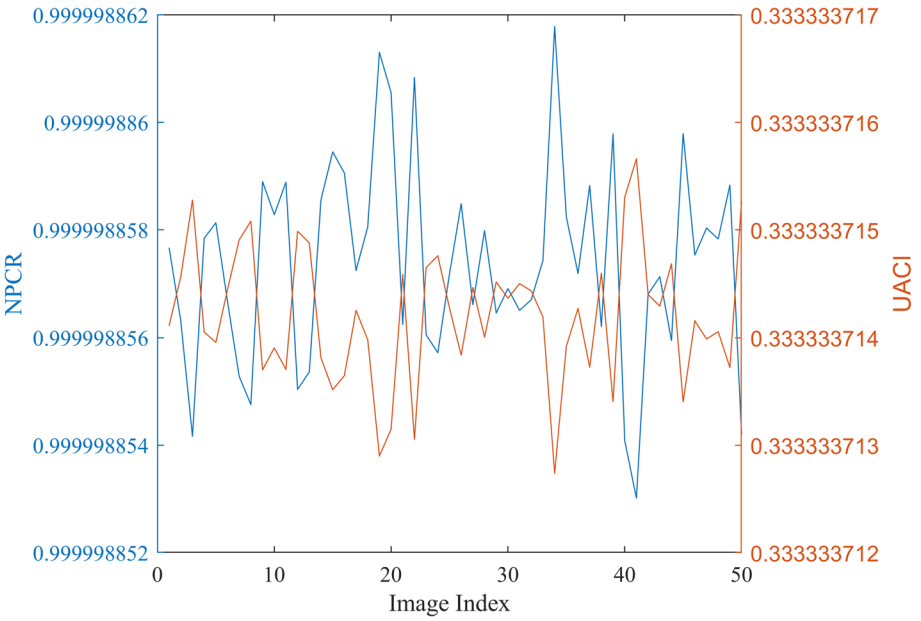


Fig. 10 The results of NPCR and UACI

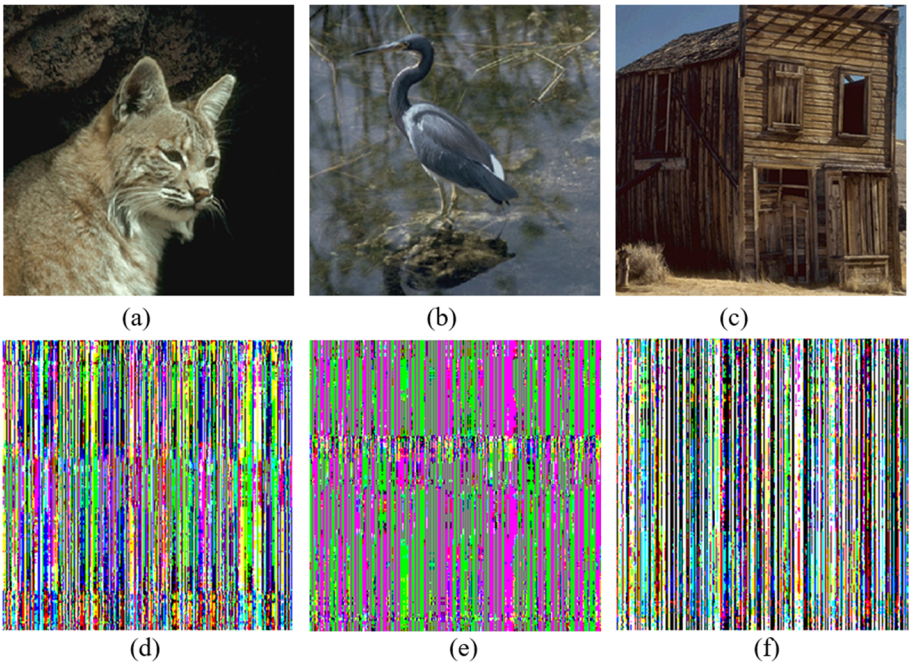


Fig. 11 The result of chosen-plaintext attack: **a**-**c** are fake plaintexts, **d**-**f** are decrypted results

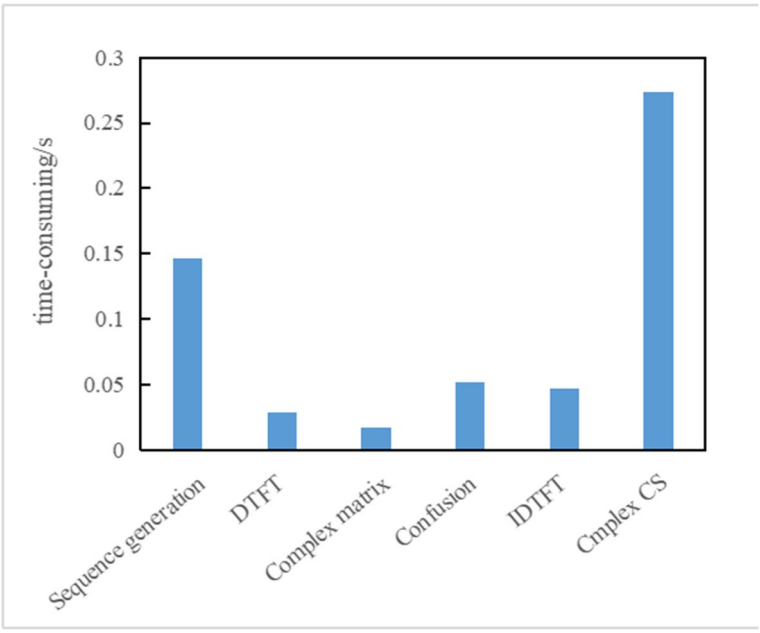


Fig. 12 Time of different modules

method [17], the CS-EMD-DFrT method [18] and the CFrFT method [21]. In order to maintain consistency, the same intensity of noise was added to all ciphertexts produced by these methods, followed by decryption using the authorized key.

Fig. 13 shows the average PSNR and CC values, it can be observed that with the increase of noise intensity, the quality of the decrypted image is degraded gradually, but the proposed algorithm consistently exhibits higher PSNR and CC values. This advantage is mainly due to the fact that the averaging operation can reduce the intensity of noise and help improve the quality of the decrypted image. Additionally, Fig. 14 displays

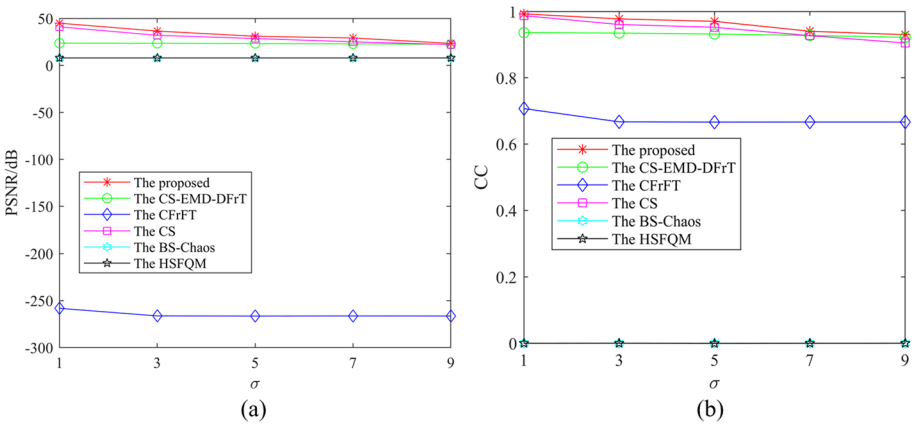


Fig. 13 The average PSNR and CC results under Gaussian noise different Gaussian noise intensities σ

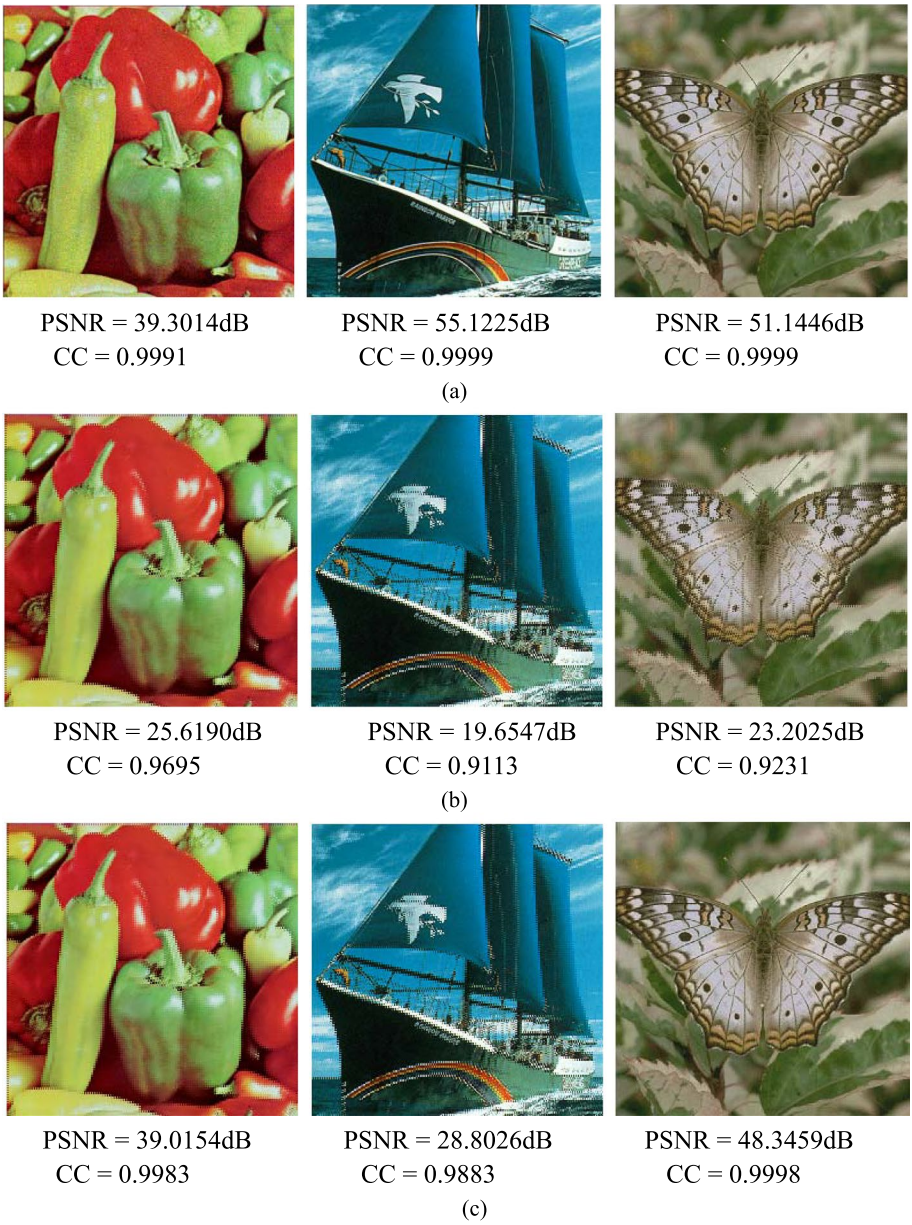
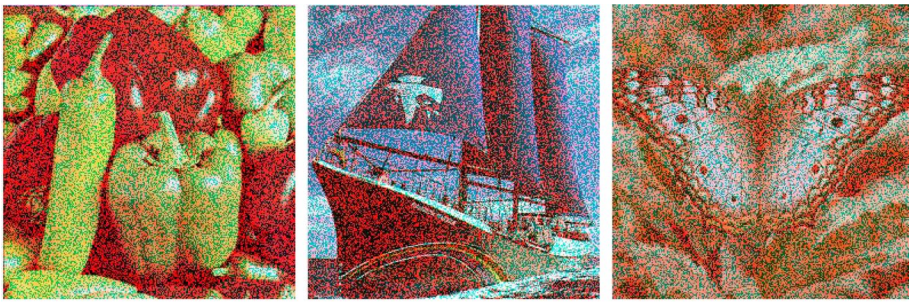


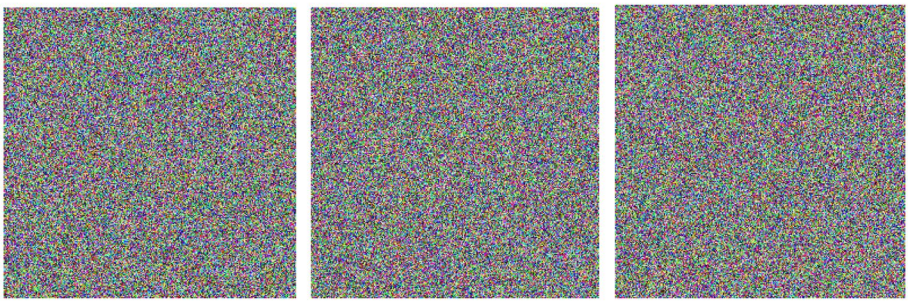
Fig. 14 The decrypted results (σ=1.0): (**a**) the proposed method, (**b**) the CS-EMD-DFrT method, (**c**) the CS method, (**d**) the CFrFT method, (**e**) The HSFQM method, (**f**) The BS-Chaos method

the decrypted results by using the six methods when the noise intensity is set at 1.0. In general, the encryption methods in frequency domain have stronger resistance to noise compared with those in the spatial domain. These results convincingly highlight the superiority of the proposed algorithm in terms of robustness against Gaussian noise.



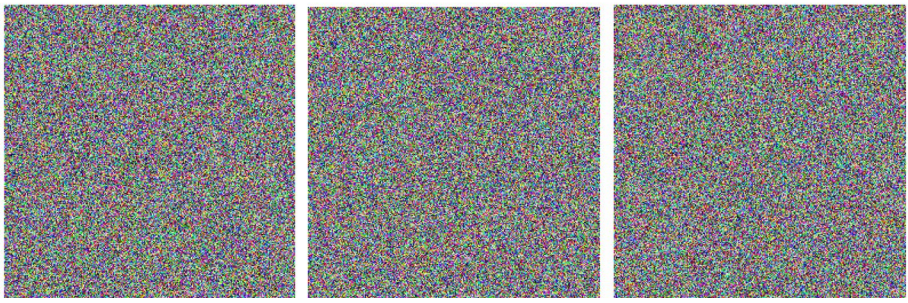
PSNR = -255.8137dB CC = 0.9797 | PSNR = -258.1242dB CC = 0.9344 | PSNR = -260.9538dB CC = 0.8013

(d)



PSNR = 8.0829dB CC = -0.0025 | PSNR = 7.3525dB CC = -0.0012 | PSNR = 9.0956dB CC = -8.8132e-04

(e)



PSNR = 8.0970dB CC = -0.0048 | PSNR = 7.3678dB CC = -0.0053 | PSNR = 9.1245dB CC = 0.0039

(f)

Fig. 14 (continued)

In addition, Table 5 lists the running times for encrypting and decrypting three color images using different methods. It is noticed that the algorithm in this paper does not have an obvious advantage in terms of time consumption, and the spatial based encryption and decryption process is faster.

Table 5 Times comparison of different methods/s

Method	The proposed	The HSFQM [6]	The BS-Chaos [10]	The CS [17]	The CS-EMD-DFrT [18]	The CFrFT [21]
Encryption	1.0584	0.1028	0.0460	0.0111	0.0243	3.9570
Decryption	0.0950	0.0233	0.0091	11.8870	24.6772	3.6556

6 Conclusion

In this paper, a novel encryption algorithm for color image is proposed. It integrates discrete trinion Fourier transform and compressive sensing. The 3D Lorenz chaotic system related with plaintext images is used to construct the measurement matrix, which solves the instability of the random measurement matrix and enhances the security of the proposed algorithm. The experimental results demonstrate the proposed encryption algorithm can recover plaintext image with high quality and better robustness against Gaussian noise compared with several existed methods. Since trinion matrix consists of three parts, the algorithm in this paper is required to process three color images at a time. In future, we will construct trinion orthogonal transform for color image and apply it to biometric template protection.

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Data availability The data are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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